

Day 3 — JavaScript Fundamentals

Programming basics + working with JavaScript in HTML / CSS (DOM manipulation)

1. What is JavaScript?

- The programming language of the web — makes pages **interactive** and **dynamic**.
- HTML = structure, CSS = style, **JavaScript = behavior**.
- Runs in the browser; also runs outside the browser (e.g. Node.js).

2. Adding JavaScript to a Page

- **Inline:** attribute on an element — `onclick="..."` (avoid).
- **Internal:** inside a `<script>` tag in the HTML.
- **External:** separate `.js` file (preferred, reusable).

```
<!-- external file, place before </body> -->
<script src="app.js"></script>
```

```
console.log("Hello from JS"); // prints to the browser console
```

3. Variables & Data Types

- **let** — value can change. **const** — value fixed. **var** — old style, avoid.
- Types: **String** "text", **Number** 42, **Boolean** true/false, **null**, **undefined**, **Array**, **Object**.

```
let name = "Abhishek"; // string
const age = 25;        // number
let isActive = true;   // boolean
typeof age;            // "number"
```

4. Operators

- Arithmetic: `+` `-` `*` `/` `%`
- Comparison: `===` `!==` `>` `<` `>=` `<=` (use `===`, checks value + type).
- Logical: `&&` (and), `||` (or), `!` (not).

5. Conditionals

```
if (age >= 18) {
  console.log("Adult");
} else {
  console.log("Minor");
}
```

```
// ternary shorthand
let status = age >= 18 ? "Adult" : "Minor";
```

6. Loops

- **for** — run a block a fixed number of times.
- **while** — run while a condition is true.
- **forEach / for...of** — loop over arrays.

```
for (let i = 0; i < 3; i++) {
  console.log(i); // 0, 1, 2
}
```

```
const fruits = ["apple", "mango"];
fruits.forEach(f => console.log(f));
```

7. Functions

- Reusable block of code. Takes **inputs (parameters)** and can **return** a value.
- Arrow functions = shorter modern syntax.

```
function add(a, b) {
  return a + b;
}
add(2, 3);           // 5

// arrow function
const greet = (name) => "Hi " + name;
```

8. Arrays & Objects

```
let colors = ["red", "green"]; // array
colors[0]; // "red"
colors.push("blue"); // add item

let user = { name: "Abhi", age: 25 }; // object
user.name; // "Abhi"
```

9. The DOM (Document Object Model)

- The browser turns your HTML into a **tree of objects** called the DOM.
- JavaScript can **read and change** this tree — that's how pages update live.
- Entry point: the global document object.

10. Selecting Elements

```
document.getElementById("title");
document.querySelector(".btn"); // first match (CSS selector)
document.querySelectorAll("li"); // all matches
```

11. Changing Content & Style

- **textContent** — change the text inside an element.
- **innerHTML** — change the HTML inside an element.
- **style** — change CSS directly. **classList** — add/remove CSS classes.

```
const box = document.querySelector(".box");
box.textContent = "Updated!";
box.style.color = "blue";
box.classList.add("active"); // toggles a CSS class
```

12. Events (Making it Interactive)

- React to user actions: **click**, **input**, **submit**, **mouseover**, etc.
- Use **addEventListener** to run a function when the event happens.

```
const btn = document.querySelector("#myBtn");
btn.addEventListener("click", () => {
  alert("Button clicked!");
});
```

13. Putting It Together (HTML + CSS + JS)

A counter that increases when the button is clicked:

```
<!-- HTML -->
<p id="count">0</p>
<button id="add">+1</button>

/* CSS */
#count { font-size: 2rem; color: #2563eb; }

// JS
let n = 0;
```

```
const out = document.getElementById("count");
document.getElementById("add")
  .addEventListener("click", () => {
    n++;
    out.textContent = n;
  });
```

Quick recap: variables store data • loops repeat • functions reuse logic • the DOM lets JS select, change, and react to elements • events connect user actions to your code.